

MASCARA GHRIS, Algeria

WMO: 605070

Lat: 35.208N

Lon: 0.147E

Elev: 1686

StdP: 13.82

Time Zone: 1.00 (E01)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	31.6	33.9	25.3	20.5	52.0	28.4	23.8	46.1	31.7	51.6	27.3	50.7	1.3	160	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	30.6	104.0	69.4	100.8	69.5	98.3	69.0	75.0	92.9	73.3	90.5	72.0	89.1	11.8	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.1	118.1	81.5	68.4	111.0	79.7	66.9	105.3	79.1	39.6	93.0	38.0	90.5	36.7	89.2	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.3	20.1	17.4	DB	27.4	109.7	2.9	1.9	25.3	111.1	23.6	112.2	22.0	113.2	19.9	114.6
			WB	26.9	78.5	3.1	1.7	24.6	79.8	22.8	80.8	21.1	81.7	18.9	83.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	64.3	48.7	49.7	54.5	59.4	67.3	75.8	83.0	82.4	74.6	67.5	56.3	50.9
	DBStd	13.34	4.81	5.70	5.59	6.16	6.60	6.53	4.54	4.52	5.06	5.95	6.01	5.36
	HDD50	239	83	67	24	3	0	0	0	0	0	0	11	51
	HDD65	2252	506	428	326	189	51	2	0	0	1	40	271	438
	CDD50	5446	42	59	164	283	535	774	1024	1006	737	543	200	79
	CDD65	1984	0	0	1	20	121	326	559	541	288	117	10	1
	CDH74	25148	1	7	67	423	1754	4209	7491	6878	2925	1281	101	11
	CDH80	13868	0	0	12	123	774	2309	4653	4141	1364	473	19	0
Wind	WSAvg	6.0	6.1	7.1	6.3	5.9	6.5	6.2	6.1	5.7	5.5	5.3	6.2	5.5
Precipitation	PrecAvg	13.8	1.7	1.5	1.4	1.7	1.1	0.4	0.2	0.3	0.8	1.3	2.0	1.4
	PrecMax	22.0	4.1	4.5	6.8	5.8	3.7	2.9	0.8	1.8	2.2	4.8	6.1	3.3
	PrecMin	7.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.5	0.0
	PrecStd	3.6	1.1	1.0	1.4	1.3	0.8	0.7	0.2	0.4	0.7	1.1	1.2	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.5	73.6	80.5	89.4	98.6	104.5	109.0	107.8	100.1	93.8	82.8	75.0
		MCWB	54.1	57.7	58.7	64.8	68.8	67.2	69.4	69.9	68.4	63.5	59.1	58.0
	2%	DB	66.5	68.2	75.0	83.0	92.8	100.2	104.4	103.7	95.3	89.4	76.6	68.4
		MCWB	52.4	54.4	58.2	61.8	65.8	67.9	69.3	70.5	67.7	63.5	57.5	54.5
	5%	DB	62.4	64.7	70.8	78.9	88.0	96.4	101.7	100.5	91.6	85.4	71.9	64.6
		MCWB	51.2	53.1	57.0	60.6	64.2	67.9	69.5	69.9	67.2	62.9	56.7	53.3
	10%	DB	59.0	61.2	67.6	74.1	84.0	92.6	98.5	97.1	87.9	81.3	68.0	61.0
		MCWB	50.6	51.8	55.9	59.3	62.9	67.6	69.0	69.5	66.7	62.5	56.0	52.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.1	59.0	62.9	67.0	72.7	75.4	76.7	77.8	74.3	69.5	64.7	58.9
		MCDB	66.0	69.0	72.3	80.7	88.0	92.6	94.6	97.1	87.8	83.0	73.4	70.2
	2%	WB	54.3	56.2	60.6	64.1	68.4	72.2	74.5	75.3	72.1	67.0	61.5	56.3
		MCDB	62.6	65.4	70.9	78.6	84.7	91.3	93.2	93.4	84.8	79.7	70.0	65.2
	5%	WB	52.8	54.4	58.5	62.0	66.4	70.3	72.9	73.6	70.6	65.4	59.2	54.4
		MCDB	60.4	62.4	68.3	75.7	81.6	89.6	91.7	90.6	83.4	78.0	67.8	62.6
	10%	WB	51.0	52.8	56.6	60.1	64.5	68.5	71.5	72.1	69.2	64.0	57.1	52.8
		MCDB	57.7	60.2	65.4	71.6	80.3	86.9	91.1	88.3	82.1	76.5	65.9	60.0
Mean Daily Temperature Range	5% DB	MDBR	20.3	20.3	22.6	24.6	27.4	29.2	30.6	29.5	26.0	24.7	20.1	19.3
		MCDBR	27.7	26.0	30.4	33.7	34.9	33.6	34.3	33.1	30.9	30.6	26.7	25.4
		MCWBR	15.5	14.7	16.6	15.7	13.9	11.4	11.8	11.7	11.3	11.6	12.6	14.0
	5% WB	MCDBR	22.7	22.8	26.8	30.0	29.9	30.0	30.1	28.8	25.9	25.4	21.9	22.0
		MCWBR	13.7	13.8	15.8	15.5	14.0	12.8	12.4	11.6	10.9	10.9	12.0	13.3
Clear-Sky Solar Irradiance	taub	0.305	0.333	0.398	0.419	0.454	0.479	0.535	0.501	0.460	0.402	0.351	0.315	
	taud	2.484	2.401	2.188	2.151	2.078	2.011	1.863	1.956	2.075	2.229	2.379	2.477	
	Ebn at Noon	286	288	275	275	266	258	243	248	252	258	264	274	
	Edh at Noon	27	32	44	48	52	56	64	57	49	38	30	25	
All-Sky Solar Radiation	RadAvg	885	1151	1550	1873	2150	2419	2384	2154	1714	1338	936	794	
	RadStd	84	117	96	148	146	81	49	77	87	89	74	62	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (12 neighbors)		N/A	N/A	N/A	N/A	+0.52	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page